

Units and Measurement

1. PHYSICAL QUANTITY

- Any property of a body that can be measured is called a **physical quantity**.
- Examples** : length, mass, time, temperature, electric current etc.

Physics is all about measuring the universe precisely. ★

2. UNITS

- A **unit** is a definite size or magnitude which is chosen as a **standard** for measurement.
- Examples** : metre (m) for length, kilogram (kg) for mass, second (s) for time.

Why do we need units?

- To measure accurately
- To compare quantities
- To communicate results universally

3. SI BASE UNITS

Base Quantity	Unit	Symbol
Length	metre	m
Mass	kilogram	kg
Time	second	s
Electric current	ampere	A
Thermodynamic temperature	kelvin	K
Amount of substance	mole	mol
Luminous intensity	candela	cd

3D Representation



← Length (m) →



Mass (kg)



Time (s)



Temperature (K)



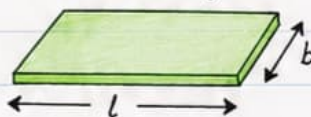
Luminous Intensity (cd)

4. TYPES OF PHYSICAL QUANTITIES

- Fundamental (Base) Quantities** :
These are independent and cannot be expressed in terms of other quantities. (e.g. length, mass, time)
- Derived Quantities** :
These are expressed in terms of base quantities. (e.g. velocity, acceleration, force)

5. EXAMPLES

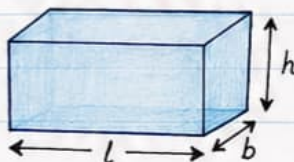
- Area of rectangle



$$A = l \times b$$

[Unit = m²]

- Volume of cuboid



$$V = l \times b \times h$$

[Unit = m³]

Key Point : A physical quantity = Numerical value × Unit



6. SYSTEM OF UNITS ★

• A set of units of all physical quantities is called a **system of units**.

• There are two systems of units.

① CGS System (Centimetre - Gram - Second)

② SI System (International System of Units)

SI system is the most widely used system in the world. ♥

Comparison of CGS and SI System

Feature	CGS System	SI System
Base unit of length	centimetre (cm)	metre (m)
Base unit of mass	gram (g)	kilogram (kg)
Base unit of time	second (s)	second (s)
No. of base units	3	7
Advantages	Simple	Universal, Accurate, Coherent

7. BASE AND DERIVED QUANTITIES

Base Quantities : The quantities which are independent of each other are called base quantities.

→ SI base quantities (7) :

- ① Length - metre (m)
- ② Mass - kilogram (kg)
- ③ Time - second (s)
- ④ Electric current - ampere (A)
- ⑤ Thermodynamic temperature - kelvin (K)
- ⑥ Amount of substance - mole (mol)
- ⑦ Luminous intensity - candela (cd)



Derived Quantities : The quantities which are obtained by mathematical combination of base quantities are called derived quantities.

Examples : Area, Volume, Velocity, Acceleration, Force, Density etc.

8. DIMENSIONS OF PHYSICAL QUANTITIES

• Dimensions of a physical quantity show how it depends on the base quantities.

• It is represented by square brackets [].

→ **Dimensions of some quantities**

Physical Quantity	Formula	Dimensions
Length (l)	l	[L]
Mass (m)	m	[M]
Time (t)	t	[T]
Velocity (v)	l/t	[L T ⁻¹]
Acceleration (a)	l/t^2	[L T ⁻²]
Force (F)	ma	[M L T ⁻²]
Energy / Work (E)	$F \times l$	[M L ² T ⁻²]
Pressure (P)	$\frac{F}{A}$	[M L ⁻¹ T ⁻²]
Density (ρ)	$\frac{m}{V}$	[M L ⁻³]

Note :

L → Length
M → Mass
T → Time



9. MEASUREMENT

• Measurement is the process of comparing an unknown quantity with a known standard quantity.

• It has two parts :

- ① Numerical value (Number)
- ② Unit

Example :

The length of a table = $\frac{1.25}{\text{Numerical value}} \frac{\text{m}}{\text{Unit}}$

10. ACCURACY, PRECISION AND ERROR

• **Accuracy :** How close a measurement is to the true value.

• **Precision :** How close repeated measurements are to each other.

• **Error :** The difference between the measured value and the true value.

High Accuracy
High Precision



Very good ✓

High Accuracy
Low Precision



Not precise ✗

Low Accuracy
Low Precision



Not good ✗

Always try to increase both accuracy and precision. ♥

★ Real Life Example :

Speedometer of a car measures speed in km/h (SI unit) which is derived from base units metre and second.



$$\text{Speed} = \frac{\text{Distance}}{\text{Time}}$$

$$[\text{Unit}] = \text{m/s}$$

11. Significant Figures

- The digits in a measurement which are known with certainty plus the first uncertain digit are called **significant figures**.
- All non-zero digits are significant.
- Zeros between non-zero digits are significant.
- Leading zeros are not significant.
- Trailing zeros are significant only if a decimal point is present.

Examples :

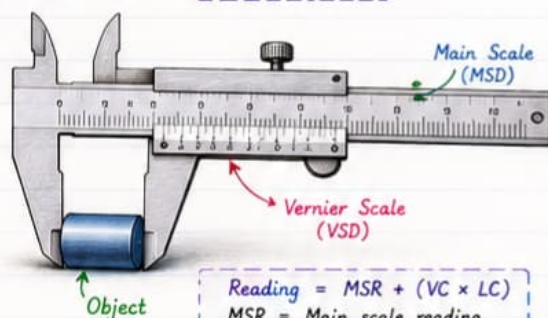
Measurement	Significant Figures	Reason
23.7	3	All non-zero digits
0.0045	2	Leading zeros not significant
1.003	4	Zero between non-zero digits
5.00	3	Trailing zeros after decimal
2400	2	No decimal, trailing zeros not significant
2400.	4	Decimal present, zeros are significant

12. Measurement of Length

- The SI unit of length is **metre (m)**.
- Commonly used instruments : metre scale, vernier calipers, micrometer screw gauge.

(A) Vernier Calipers

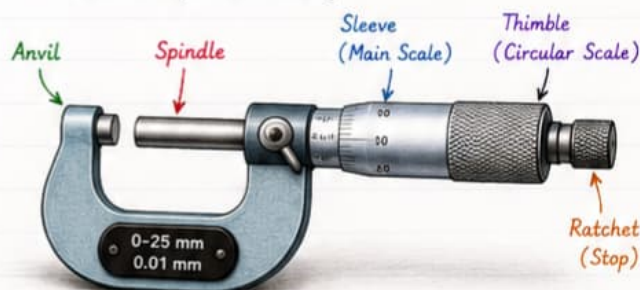
- Used to measure external diameter, internal diameter and depth.
- Least count = $1 \text{ MSD} - 1 \text{ VSD}$



Reading = $\text{MSR} + (\text{VC} \times \text{LC})$
 MSR = Main scale reading
 VC = Vernier coincidence
 LC = Least count

(B) Micrometer Screw Gauge

- Used to measure very small lengths (thickness/diameter of thin wire, sheet etc.).
- Least count = $\frac{\text{Pitch}}{\text{No. of divisions}}$
 $= 0.01 \text{ mm}$ (in most cases)



Reading = $\text{MSR} + (\text{CSR} \times \text{LC})$
 CSR = Circular scale reading



Always write the final answer with correct unit and proper significant figures. ✓

13. Measurement of Time

- The SI unit of time is **second (s)**.
- Instruments used : Clock, Stop watch, Digital timer, Pendulum clock.
- For short intervals → Stop watch
- For long intervals → Clock



Stop watch



Digital timer

1 min = 60 s
 1 h = 60 min = 3600 s



Common Unit Conversions (Length)

- 1 km = 1000 m
- 1 m = 100 cm = 1000 mm
- 1 cm = 10 mm
- 1 mm = 10^{-3} m

Common Unit Conversions (Mass)

- 1 kg = 1000 g
- 1 g = 1000 mg = 10^6 μg
- 1 mg = 10^{-3} g
- 1 μg = 10^{-6} g

14. Measurement of Mass

- The SI unit of mass is **kilogram (kg)**.
- Instruments used : Beam balance, Electronic balance.



Beam balance



Electronic balance

Note : Mass of an object does not change from place to place.



★ Key Takeaways

- ✓ Always use SI units for accurate communication.
- ✓ Use correct instruments for measuring physical quantities.
- ✓ Record observations carefully and report with proper units and significant figures.
- ✓ Accuracy in measurement is the first step in scientific investigation.





15. Dimensional Analysis

- Dimensional analysis is a method to check the correctness of a physical equation in terms of **dimensions**.
- The dimensions of L , M and T are taken as fundamental.
- If both sides of an equation have same dimensions, then equation is **dimensionally correct**.
- Useful to derive formula up to a dimensionless constant.

Example : Derive the formula for time period (T) of a simple pendulum.

$$\text{Let, } T \propto L^a g^b$$

$$\text{Dimensions on LHS} = [T]$$

$$\begin{aligned} \text{Dimensions on RHS} &= [L]^a [LT^{-2}]^b \\ &= L^a \times L^b T^{-2b} \\ &= L^{a+b} T^{-2b} \end{aligned}$$

Equating powers of L and T

$$\begin{cases} a + b = 0 & \dots (i) \\ -2b = 1 & \Rightarrow b = -\frac{1}{2} \end{cases}$$

$$\text{From (i)} \Rightarrow a = \frac{1}{2}$$

$$\therefore T \propto L^{1/2} g^{-1/2} \Rightarrow T = 2\pi \sqrt{\frac{L}{g}}$$



16. Errors in Measurement

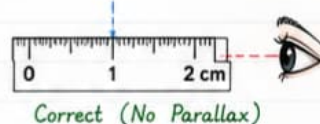
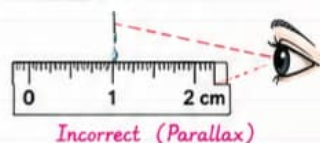
- Errors are of three types :

- ① **Systematic Errors** : These are consistent, repeatable errors that occur due to faulty instruments or wrong techniques.
- ② **Random Errors** : These occur unpredictably due to sudden changes in environment.
- ③ **Gross Errors** : These occur due to carelessness or mistakes by the observer.

Ways to Minimize Errors

- ✓ Use calibrated instruments.
- ✓ Take multiple observations and calculate mean value.
- ✓ Avoid parallax error.
- ✓ Use appropriate range of instrument.
- ✓ Use proper technique and care.
- ✓ Report result with proper significant figures.

Parallax Error



17. Estimation

- Estimation means finding the approximate value of a quantity using reasoning and experience.
- It helps in quick calculations and in checking the reasonableness of a result.

Examples (Estimation)

- Thickness of a coin $\rightarrow \approx 2 \text{ mm}$
- Length of a pencil $\rightarrow \approx 18 \text{ cm}$
- Time for one blink $\rightarrow \approx 0.1 \text{ s}$
- Height of a room $\rightarrow \approx 3 \text{ m}$
- Mass of a notebook $\rightarrow \approx 200 \text{ g}$



Important Formulas (Quick Reference)

Physical Quantity	Formula / Relation	Dimensions
Speed (v)	$v = s/t$	$[L T^{-1}]$
Acceleration (a)	$a = \Delta v/t$	$[L T^{-2}]$
Force (F)	$F = ma$	$[M L T^{-2}]$
Work (W)	$W = Fs$	$[M L^2 T^{-2}]$
Energy (E)	$E = mgh$	$[M L^2 T^{-2}]$
Pressure (P)	$P = F/A$	$[M L^{-1} T^{-2}]$
Density (ρ)	$\rho = m/V$	$[M L^{-3}]$

Solved Example

Q. A car covers 120 km in 2 hours. Find its average speed in SI units.

Sol. Distance (s) = 120 km = $120 \times 1000 \text{ m} = 120000 \text{ m}$
 Time (t) = 2 h = $2 \times 3600 \text{ s} = 7200 \text{ s}$

$$\text{Average speed, } v = \frac{s}{t} = \frac{120000}{7200} = 16.67 \text{ m/s}$$

Ans. 16.67 m/s



120 km
in
2 hours

Final Revision (One Line Revision)

- ★ Physics is the science of measurement.
- ★ Use SI units for universal understanding.
- ★ Accuracy comes from good instruments and careful observation.
- ★ Correct significant figures show the reliability of result.
- ★ Dimensional analysis checks the correctness of equations.

Key Points to Remember

- ✓ Always write unit with the numerical value.
- ✓ Never mix different systems of units.
- ✓ Keep track of significant figures.
- ✓ Report final answer with unit.
- ✓ Practice estimation to build intuition.



Measure Carefully, Think Accurately, Report Responsibly.